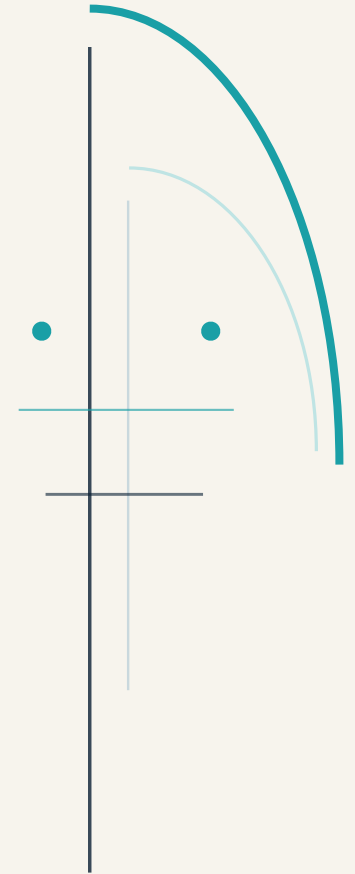


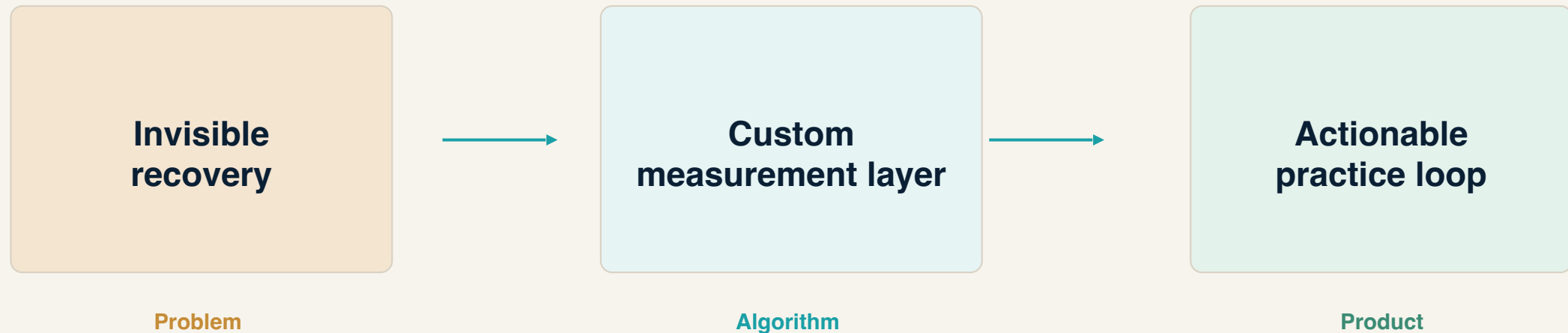
Mirror for Bell's Palsy

A local-first facial retraining companion that turns invisible recovery into measurable practice.



The executive thesis

Bell's palsy recovery depends on feedback, but the patient cannot reliably feel or see the millimetres that matter. Mirror converts webcam geometry into practice-grade signals, while keeping the product local, private, and conservative.



The patient problem is a broken feedback loop

“ *When half your face won't move, the mirror stops being useful.* ”

WHAT RECOVERY ASKS FOR

Daily, gentle, repetitive movement
Evidence that effort is changing something
Confidence that a bad capture is not “bad recovery”

Skip reps

Can't feel

Can't see

Can't trust

Without measurement, motivation becomes guesswork.

Mirror at a glance

A practice companion, not a clinician. It guides exercises, tracks webcam landmarks with MediaPipe, scores balance and affected-side movement, and helps users export useful evidence.



Guided practice

Forehead, eyes, nose, cheeks, mouth, vowels, and expression-style movements.



Real-time scoring

Live left/right symmetry plus affected-side movement against personal baselines.



Progress memory

Sessions, journals, streaks, trend charts, report images, and assessments saved locally.



Review output

Printable reports, clinician bundles, and validation exports when the user chooses.

Primary URL: mirror.iali.in · Demo: mirror.iali.in/try

Two product surfaces, one measurement core

The demo gets trust in seconds; the practice app builds longitudinal recovery evidence. Same tracker, different decision context.

shared measurement core

/TRY DEMO

Instant tracker proof

- 478-point mesh color-coded by region
- Ranked muscle activations with left/right bars
- Live expression chip and head-position readout
- No session machinery or scoring lock

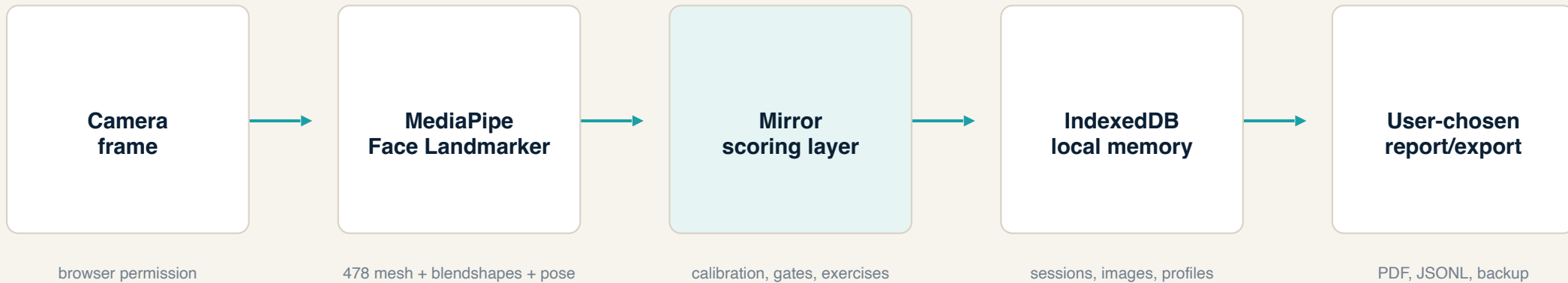
FULL PRACTICE APP

Measured recovery loop

- Camera setup → calibration → exercise hold → feedback
- Movement profile, adaptive plan, comfort dosing
- Assessment trend, journal, reports, local exports
- Unscored guided practice still works when camera/model fails

Architecture: local-first by construction

The pipeline is intentionally boring in the right places: browser camera, on-device inference, custom scoring, IndexedDB, explicit export. No account. No practice-data backend.



Worker-backed detector where supported, main-thread fallback where not.

The model does geometry; Mirror does judgment

This distinction matters. MediaPipe is the sensor; Mirror is the scoring product. Mixing those up would make the deck sound like demo-day fog machine nonsense.

MEDIAPIPE PROVIDES

478

3D landmarks

blendshapes

blink, brow, smile, sneer

pose

facial transform matrix

Loaded on-device with GPU delegate;
inference uses Face Landmarker
0.10.21 runtime.



MIRROR ADDS

Neutral calibration and noise floors
Pose-normalized face-local scoring
Exercise-specific directional logic
Affected-side progress against baselines
Reports, diagnostics, safety copy

Capture trust stack: score only when the data deserves it

Mirror is conservative by design. If evidence is weak, the better answer is silence or coaching—not a confident-looking number wearing a cheap suit.

01

Setup quality

face presence · lighting · distance · FPS · stability · glare risk

02

Neutral calibration

24 stable frames · per-landmark jitter · neutral pose matrix

03

Hold-frame gates

head-pose deviation < 0.20 rad · movement above noise

04

Scoring diagnostics

drop reasons · rejected frames · capture flags · safety notes

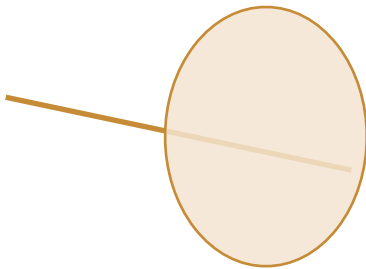
Calibration: CALIBRATION_FRAMES = 24, stability epsilon = 0.006, center offset < 0.12, eye-line tilt < 0.12 rad.

Updated normalization: measure the face, not the camera wobble

The scorer now prefers MediaPipe's 3D facial transformation matrix, converting landmarks into a face-local coordinate frame before comparing movement.

BEFORE

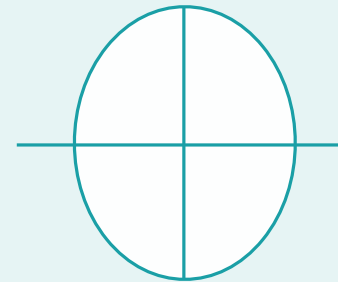
Screen-space landmarks



Head tilt, distance, and yaw can masquerade as movement.

AFTER

Face-local frame

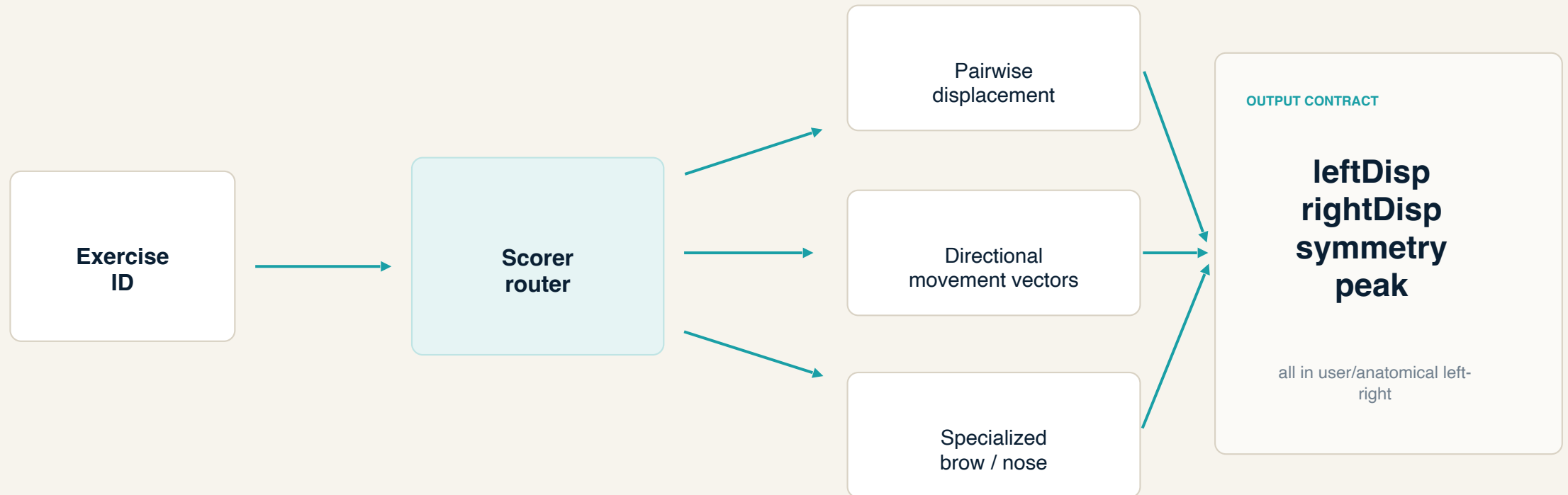


Translate to nose tip
Apply inverse pose rotation
Scale by 3D inter-ocular distance
Residual eye-line roll correction

Practical result: translation, roll, scale, and much yaw/pitch foreshortening get reduced before scoring.

Scoring engine: route each movement to the right scorer

Generic distance scoring is lazy. A brow raise, nostril flare, smile, and pucker do not move the same way. Mirror routes exercises to purpose-built scorers.



Movement families now have movement-specific logic

The updated algorithm counts the intended direction, not merely “any landmark moved.” That is how you stop wrong movement from being rewarded.

Family	Signal counted	Why it matters
Smile · vowel E/I	outward + slightly upward mouth pull	prevents pucker from scoring as smile
Pucker · kiss · vowel O/U	inward lip motion toward midline	prevents smile-only movement from scoring
Eye close · blink · wink	decreased eyelid aperture	rejects lateral eye drift
Brow raise / frown	brow-eye gap; frown down/inward	stable against absolute brow position
Nostril flare / scrunch	rim/ala aperture; upward lift + sneer blendshape	no fallback to generic if direction fails

Blendshapes help activation and snapshot timing. They are not the main symmetry score because coefficients can look artificially balanced on asymmetric faces.

The metric split: balance is not recovery

A symmetry score can stay flat while the affected side genuinely gets stronger. The updated deck makes that distinction explicit.

MOVEMENT BALANCE

$$\text{symmetry} = \min(\text{left}, \text{right}) / \max(\text{left}, \text{right})$$

Best for: “was this rep balanced?”
Not enough for: “am I recovering?”

RECOVERY PROGRESS

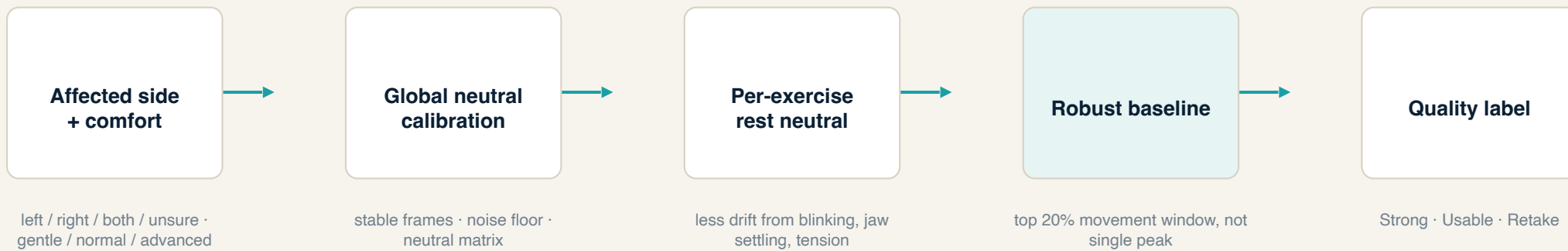
$$\text{affectedProgress} = \text{current affected} / \text{first baseline affected}$$

Also tracks affected-to-proper ratio today versus baseline.

Example: “Right smile is +18% from first baseline. Affected-vs-proper is 62% today, up from 48%.”

Personal movement profile: baseline as a living layer

The profile is not a trained model. It is a local, individualized baseline and threshold system generated from the user's own movement.



Threshold bands: minimumVisible = 20% of peak floor 0.002 · reliableMovement = 35% of peak floor 0.004 · baselineTarget = max peak

Nose baselines use lower nose-specific thresholds because nostril flare is typically only 1–2% of inter-ocular distance.

Adaptive practice: recommendations with guardrails

The app prioritizes useful practice without pretending noisy scores are insight. This is the part where “move fast” would have shipped dangerous nonsense.



Standard assessments make progress reviewable

Daily practice and standardized assessment should not be mashed together. Mirror stores assessment records separately so trends can be compared cleanly.

FIXED MOVEMENT SET

- 01 Eyebrow raise
- 02 Gentle eye closure
- 03 Open smile
- 04 Nose wrinkle / snarl
- 05 Lip pucker

STORED RECORD

- Rest metrics** palpebral fissure · midface proxy · mouth-corner height
- Voluntary movement** brow / eye / midface-nose / mouth zones
- Coactivation risk** quiet-region movement during intended movements
- Report linkage** source session images and printable report

Assessment sessions do not count toward daily practice goals or streaks.

Clinical-scale estimates: useful, fenced, not clinical grades

Dedicated caveat, because this is where sloppy decks get people in trouble. Mirror can show scale-inspired self-tracking estimates, but it does not diagnose, grade paralysis, or replace clinician assessment.

DISPLAY EVIDENCE GATE

60%

usable standard-assessment
movement coverage

plus complete resting asymmetry metrics and
usable/strong capture quality where present.

ESTIMATES PRODUCED

House-Brackmann-inspired global
estimate
Sunnybrook-style composite estimate
eFACE-style static / dynamic / synkinesis
proxies

v5: partial estimates omit weak/missing
movements instead of treating them as zero.

VALIDATION RELEASE GATE

80%

observed agreement + Wilson
lower bound

Requires reviewed clinical labels, 30
assessments, 10 distinct cases, source hashes,
reviewer controls, and status update.

Current status: `clinicalFacingScoresAllowed = false`. The app remains estimate-only until reviewed evidence exists.

Reports and exports turn practice into reviewable evidence

The user controls every export. Mirror packages context, quality, and trend evidence so a clinician is not staring at a naked score with no provenance.

EXPORT 1

Printable report

Session date, comfort level, setup quality, symmetry, per-rep chips, affected-side progress, baseline/current images, safety notes.

EXPORT 2

Clinician bundle

JSONL review package with assessments, comparisons, sessions, journal notes, quality flags, selected images, frame samples.

EXPORT 3

Validation dataset

Opt-in JSONL with frame samples, assessment scale rows, label templates, source context, and replay metadata.

Backup import/export is separate from clinician/validation JSONL. That separation is boring and exactly why the data model is not a junk drawer.

Privacy posture: local-first, explicit-share only

A face-tracking product only earns trust if the privacy model is not a footnote. Mirror's practice data stays in the browser unless the user exports it.

STAYS LOCAL

- Camera frames processed in browser by MediaPipe
- No uploaded video or Mirror practice-data backend
- Sessions, landmarks, profiles, journal, images in IndexedDB
- Optional local recovery model trains from saved device data

LEAVES ONLY BY USER CHOICE

- Print/save report as PDF
- Export clinician bundle JSONL
- Export validation dataset JSONL
- Clear site data to remove local Mirror data

Hosting caveat: Render may process normal hosting metadata. Mirror's code does not use it for practice-data collection or profiling.

Validation framework: tooling ready, clinical evidence pending

The repo now has a release-gated validation workflow. The honest status is stronger than hand-waving: tooling exists; reviewed clinical datasets do not.

CURRENT STATUS

**tooling-ready
needs-reviewed-data**

Reviewed datasets



Production threshold calibration



IMPLEMENTED WORKFLOW

Validation JSONL export
Blinded clinical review package
Label merge + reviewer agreement
Replay evaluation + threshold calibration
Clinical-readiness + release status checks

FUTURE RELEASE GATES

30 reviewed assessments
10 distinct validation cases
80% observed + Wilson lower bound
HB case-mix floors
Source SHA-256 traceability

Technology stack: modern, local, lightweight

The stack is intentionally client-heavy: fast to ship, easy to host, and aligned with the privacy promise.



Also: Recharts for progress charts · Lucide icons · WebAssembly / WebGL · worker-backed detector with fallback · OpenAI Codex, Claude Opus, and Pieces in the build workflow

Roadmap: where the next credibility comes from

Most core algorithm upgrades are implemented. The next value is not more sparkle; it is validation, threshold calibration, and cleaner review workflows.

TRACK 1

Now implemented

- Worker-backed inference
- Directional scorers
- Setup quality + diagnostics
- Assessment mode + exports
- Validation label workflow

TRACK 2

Pending evidence

- Clinician-reviewed datasets
- Production threshold calibration from reviewed reports
- Clinical-scale agreement proof
- Reviewer-agreement proof

TRACK 3

Product polish

- Portable progress import/export
- Clearer calibration coaching
- Better clinician handoff summaries
- More replayable audit trails

Design principle: do not make the app look clinically “official” until the evidence is actually official. Restraint is the feature.

Distribution: the story already has pull

The origin story landed because it is specific: one person, a sudden diagnosis, and a working tool built out of real need.

9+

publications picked up the story

May 2026

media attention window

local-first

privacy narrative, not just tech demo

Featured across Hindustan Times, Times of India, Mint, Free Press Journal, NewsBytes, What's Hot, MSN, LiveMint, Dailyhunt and others.

Executive framing: press proves resonance, not clinical validity. Keep those lanes clean.

Try it yourself

The executive story is simple: see the tracker immediately, then understand the full practice loop.

SEE THE TRACKER

mirror.iali.in/try

Standalone face mesh, activation bars, expression detector, head-position readout.

FULL PRACTICE APP

mirror.iali.in

Guided practice, calibration, baseline profile, trends, reports, and exports.

Medical caveat travels with the product: practice and self-tracking only; work with a qualified clinician.

CLOSING

From invisible recovery to measurable practice.

Mirror does not promise recovery. It gives people a private, repeatable way to see whether their practice is changing over time.

BUILT BY

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SUPPORTED BY

OpenAI · Render · MediaPipe · Claude · Pieces

mirror.iali.in